LIFTING A SEPARABLE RESIDUE FIELD INTO A COMPLETE LOCAL ALGEBRA

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Separable field extensions are formally smooth, and formal smoothness upgrades to the adic topology. Algebra, Proposition \ref{algebra-proposition-characterize-separable-field-extensions} and Lemma \ref{lemma-formally-smooth}, cited at proof lines 9598-9601. Give formal smoothness of the residue field extension in the required sense.
- Existence of a lift of a continuous map into a complete local ring along the quotient. Lemma \ref{lemma-lift-continuous}, cited at proof lines 9602-9603. Produces the section of the quotient map.

Source reference. Original lemma 0C34: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15179–15179. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0322: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4005–4005. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07EC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9442–9442. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 07EB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9441–9441. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07NJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9764–9764. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let k be a field. Let $(A, \mathfrak{m}, \kappa)$ be a complete local k-algebra. If κ/k is separable, then there exists a k-algebra map $\kappa \to A$ such that $\kappa \to A \to \kappa$ is id_κ.*

Proof. By Algebra, Proposition 0322 the extension κ/k is formally smooth. By Lemma 07EC $k \to \kappa$ is formally smooth in the sense of Definition 07EB. Then we get $\kappa \to A$ from Lemma 07NJ. □